

Institute of Actuaries of India

Subject CP1 – Actuarial Practice (Paper A)

June 2019 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

The features that can increase the risks in a product design are:

- lack of historical data
- high guarantees
- more options
- overhead costs
- complexity of design
- untested market

[3]

Solution 2:

i) Motor insurance – the analysis of data obtained from using telematics (i.e. monitoring driver behaviour through a device installed in the insured vehicle).

- Duration of trips in pricing a product i.e. the more the trips, the higher the premium [1]
- Cars connected with GPS can be used in finding the exact location of claims [½]
- Breaking and acceleration used by the drivers can decide the premium [½]

[2]

ii) Health insurance – the analysis of exercise regime (e.g. from wearable fitness monitors).

- Wearables like fitness watches can be used in tracking the activities by the policyholders eg. no of steps, heart rate, BP, other forms of activity etc
- Points for doing exercise and reduce future premiums

[2]

[4 Marks]

Solution 3:

i) Global provision is reserves calculated at portfolio or line of business level to estimate the risk that are not taken into account while calculating per policy reserves. [1]

For example, an additional provision for mismatching of assets and liabilities, IBNR reserves, etc. [1]

[2]

ii) Global provisions are kept due to following reasons:

- May be regulatory requirement [½]
- To act as additional protection against insolvency [½]
- To cover risks, both financial and non-financial, that cannot necessarily be attributed to individual contracts [1]

[2]

[4 Marks]

Solution 4:

Protection of employees- [1]

- Life Insurance cover for workers in case of any accidental death at the plant/site
- Health Insurance cover for workers in case of any accident at the plant for treatment in a hospital

Property Damage Insurance: [1]

- This is insurance to indemnify against loss of or damage to vehicles, machinery and equipment etc used at oil drilling site, due explosion or other reasons

- Business interruption cover- Collapse of operations due to NAT CAT events

Liability Insurance: [1]

- Third party public liability (environmental pollution caused),
- Professional indemnity-errors and omission for employees and directors

Marine insurance: [1]

- Large individual claims – for example, the loss of a large ship at sea
- Accumulations – for example, a storm hits a harbour and equipment and machinery for transport are damaged.

Financial Loss insurance: The benefit provided by financial loss insurance is indemnity against financial losses arising from a peril covered by the policy. [1]

It includes:

- pecuniary loss – perils include bad debts or failure of third parties, includes mortgage indemnity guarantee insurance
- fidelity guarantee – To cover against dishonest actions by employees, such as fraud and embezzlement
- business interruption e.g. political reasons

[5 Marks]

Solution 5:

- The company will probably prefer to continue with the existing plant unless the price being offered by the developer is significantly above both the market price of the facility and the costs of uprooting and rebuilding the manufacturing facility elsewhere and this being the case here, he may recommend it to the Shareholders. [1]
- This place is already being developed as residential place so this may be the right time and opportunity to take this up to the Shareholders to gain maximum return out of sale. [½]
- Besides the above, the other general points he needs to consider in his mind are:
 - The 'market price' of the facility could be estimated by looking at what other similar properties have been sold for in recent times. [½]
 - If the company simply wants access to fresh finance, it could obtain this by selling the property to a property investor and then leasing it back. [1]
- The price offered must reflect:
 - The location, in particular transport links and proximity to workforce, customers and suppliers [½]
 - The age and condition of the building [½]
 - Whether it can be adapted to other uses [½]
 - The number of comparable facilities [½]
 - The lease structure [½]
 - The size of the plant. [½]
- The price offered would also have to be greater than the costs of relocating if required. [½]
- These would include:

- The costs of laying-off local staff and re-hiring new staff in a new location [½]
- The cost of searching for a new suitable location which has all the necessary features of transport and communication links.

- Other implications such as taxable gains would also need to be considered. [½]
- As well as considering the price offered, there will be non-financial factors that must be considered. [½]
- Does the company have an agreement with the local authority that restricts it from selling its plant to the developer? [½]
- Will public opinion swing against the car company if it sells a site to developers, particularly if redundancies are involved in the local market? [½]

[6 Marks]

Solution 6:

Adapting existing models

Merits

- The company will not incur the cost of buying a new model, but will need to bear the costs of adapting existing models.
- Existing staff will have an understanding of the workings and limitations of the existing models
- existing models should be well-documented
- By adapting an 'in-house' model, existing staff can ensure consistency of design with existing models.
- Existing staff will have developed expertise in modelling, reducing future reliance on external providers for modifications.
- Purchasing a new model may be more expensive than upgrading existing model
- The external model may not be well tested.
- It should be well designed and validated, in order to be a successful commercially –
- Look at testimonials, other users, success in other markets in RBC
- The current software may be more flexible than and therefore may be more useful in future for other purposes (eg valuation, ALM, projections).
- New software means and new vendor Feeling of loss of control over development and
- Costs of Training of resources on new software,
- Timing for implementation of both- regulatory time constraints in implementing RBC-
- external vendor may not be able to implement

[7 Marks]

Solution 7:

i) When using competitor data for comparison, the company should consider differences in:

- The scope of the cover provided, ie types and levels of benefits i.e. he needs to consider the indemnity Vs Fixed Health Benefits products as both offers different kind of types and level of benefits e.g. indemnity provides the cost associated with the Hospitalisation while the fixed benefit provides the fixed cost irrespective of the event
- The target market (expected claims experience may vary significantly by the types of policyholder targeted) of the company and its competitors
- Business management actions taken such as underwriting levels and

- Use of reinsurance the approach taken to valuation by the different companies,
- Level of differences due to choice in how prudent the basis is
- The quantity of the data (ie number of policies) and hence how credible it is which would indicate reliance based on own experience.
- The company should also consider how reliable the source of data has proved to be in the past, and when the results were compiled (they may be out-of-date) [4]

ii) The major areas shall be:

Data used for valuation

[2]

1. First section shall be on the data used for statutory valuation and the data validation process to validate that the data is accurate and complete.
2. Discuss Data extraction from core system and adjustments if any
3. List all the checks that have been performed on the data e.g.
 - Number of records comparisons with last month data
 - Amount of new business recorded by internal MIS team versus new business premium in the data received
 - Date formats must be in order like Date of birth
 - Comparison of sum insured with last month's data etc
4. Ratio/trend Analysis done on the data is also important e.g. average sum insured, average premium, Premium / 1000 sum insured etc.
5. Process for documentation of data errors and action to be taken shall be recorded.

Methodology

[2]

6. This section shall cover all types of reserves that the company is holding:
 - Unexpired Risk
 - IBNR Reserves
 - Asset liability mismatching reserve
 - Catastrophe reserves
7. The methodology used to determine these reserves shall be written clearly.
8. All relevant regulations/professional guidelines shall be referred in this section.
9. Treatment of reinsurance should be documented

Experience Investigation for eg:

[2]

10. Should discuss internal/external analysis to be conducted before setting parameter values
11. This includes analysis of claims data, expense data, asset data etc. To set incidence rates, expense assumption and discount rate for valuation purpose.
12. The granular level to which Claims data may need to be analysed -at product, sales channel, duration of entry, smoker/non-smoker status, age and gender level to determine incidence rates, should be documented.
13. Process for doing trend analysis for eg., Is arrived incidence rates are better/worse compared to last year claims analysis and reasons for significant deviation?
14. Then arrive at incidence rates to be used for valuation purpose (after allowing for margin of adverse deviation)
15. Process for performance of expense investigation must be documented-- expenses will need to be analysed to determine expense loadings for provisions. Try to do initial and renewal expense allocation and arrive at per policy unit cost.

16. Documentation on process to be adopted for inflating expenses for the future should be discussed
17. What are the relevant regulatory guidelines and practice standard that need to be considered for setting assumptions for statutory valuation?

Result analysis

[1]

18. This section includes are major checks which shall be done on final reserve numbers. E.g.
- What is reason for changes in average trends in reserves, seasonality impact etc.
 - Product level reserves shall be compared with last year/ month reserve figures

[7]

[11 Marks]**Solution 8:**

i) The reasons behind the Chief Actuary's recommendation may be:

- Given that, this being a new product the Actuary may want a reduction in claims volatility and hence reinsurance is very much required [½]
- XoL arrangements will protect from Large and one-off claims and to limit the amount of large losses arising from:
 - A single claim on a single risk [½]
 - A single event [½]
 - Cumulative events [½]
 - portfolio concentrations of risk [½]
- The support from the reinsurer through XoL agreements will smoothen the profits and [½]
- Quota share will reduce the capital requirements. [½]
- Given that, a high volume is expected due to very competitive pricing, quota share will give an increased capacity to write more business and achieve diversification [½]
- Given that this product being new, technical inputs to pricing/underwriting may be required [½]
- And last but not the least, this being a new product line, the regulator may require an adequate reinsurance in place for stability [½]

[5]

ii) Assessing the security of a reinsurer

- The Finance Director's concern is genuine as the insurer is ultimately responsible for the risks it carries and hence one should be careful in getting the reinsurer on board. [1]
- But, it is also necessary to pass on some risk to reinsurer so that Company's profits and solvency stay stable especially the product is new and the Company has no past experience. [½]
- The reservations can be addressed - it is important to analyse credit worthiness of reinsurer before any decision is taken. [½]

[2]

iii) The following characteristics may be checked while taking decisions on the same:

- The insurer should start by looking at the financial strength of the reinsurer and this may be checked by the ratings given by the credit rating agencies. [½]
- The reinsurer's published accounts and statutory returns would be a good starting point. [1]
- The insurer should look at the reinsurer's:
 - Solvency position, i.e. the level of free assets [½]
 - Types of assets held – match for liabilities, cash flow, volatility, diversification, marketability and liquidity [1]

- Borrowings – level of gearing, income and asset cover. [1]
- The legislative environment in which the reinsurer operates will also affect solvency. [½]
- The type of business that the reinsurer writes will affect the security of the reinsurer. [½]
- With regard to the reinsurer's liabilities, the insurer should look at:
 - the diversity by class, geographical region and cedant [½]
 - exposures to any risk accumulations / catastrophes [½]
 - any particularly large, unusual or volatile risks covered by the reinsurer [½]
 - the size of the liabilities and future plans for development / expansion [½]
 - whether the reinsurer itself has reinsurance. [½]
- The insurer should examine the reputation of the reinsurer, for example:
 - its market share [½]
 - its previous claims history [½]
 - the ability of the management [½]
 - whether the reinsurer has a parent company or is associated with other companies and
 - how financially strong these are [½]
 - its credit rating [½]
 - the views of others in the industry: other insurers, analysts, the regulator etc. [½]

[5]

[12 Marks]

Solution 9:

i) In this option, the Company can invest on behalf of their customers. The following are additional risks:

- Risk of earning lower return on the portfolio than expected by customers
- Cost of doing investments e.g. custodianship fee, salary to investment managers etc. can this cost can be recovered from the customers?
- Reputational risks if the returns are lower than competitors
- Additional regulatory compliances on investment portfolio
- Default risk of any security...reputational loss?
- Liquidity risk from issuer

[3]

ii)

- Profile of the liabilities is the key consideration to set the investments. In the early period for a customer, a part of the investments can be done in equities with aim to earn higher returns so that corpus grows.
- During last 5-7 years before retirement for a customer, his investments shall be gradually moved into more secure assets like AAA rated corporate bonds, government bonds etc.
- Diversification in terms of asset class, issuer, industry, group companies etc. shall be maintained.
- The investment options offered must take into account legislative restrictions.
- Keep some asset in liquid forms like fixed deposits, cash etc. to match expenses, benefit payouts in case of death etc.
- Profile of target market – consider the financial sophistication of the company's customers in making investment decisions.
- Hence, the investor's risk appetite must be considered.
- The level of risk aversion of the target market is critical, as it will determine the extent to which risk (including mismatching risk) may be taken in pursuit of higher returns.

- The tax treatment of the different asset classes should be considered.
- Practical issues – the size of the funds under management might limit the range of available asset classes.
- Frequency of switching – limits may be imposed on how often members can amend their investment choice (eg between the lifestyle option and other self-select choices) to reduce administration costs.
- The expense of investing in any particular asset class shall be considered.
- Competition – consider the comparable investment options offered by competitors. [6]

iii)

- The term risk budgeting refers to the process of establishing how much risk should be taken and where it is most efficient to take the risk in order to maximise return.
- The insurance company will need to determine the overall risk budget, allowing for strategic, structural, and active risks.
- It will then need to determine how it will allocate these risks within the strategy.
For example,
 - whether high quality bonds or just government bonds
 - or what types of corporate bonds will be permitted
- If corporate bonds will be permitted, a credit risk budget will be needed for the matching portfolio
- The active risk budget will only be relevant for the equity portfolio.
- Currency risk will also need to be budgeted within the equity portfolio.
- Need modelling of the equity portfolio to assess risk of future variance of return compared with benchmark.
- If a number of different asset managers are used within each portfolio the risk budget will need to be divided between them.
- Any diversification benefits from different asset managers would also need to be allowed for in the risk budget.
- The level of free assets shall be considered while doing this exercise.
- Further, shareholder or policy holder expectations must be considered before finalising risks budgeting. [5]

[14 Marks]

Solution 10:

i)

- We have to consider total expenses incurred by the Company apart from commissions for this purpose.
- Since, the Company is writing majorly non-participating endowment business and hence, primarily expenses related to this line of business.
- Recent expense figures must be picked up for analysis.
- The first step is to allocate all expenses, function / department wise e.g. Finance, HR, IT, Sales etc.
- We must separate out any one-off expenses like purchase of new IT system etc.
- The expenses can be direct as well as indirect.
- The allocation of direct expenses is straight forward. However, allocating indirect expenses are tricky.
- The allocation of indirect expenses shall be done on nature of expenses and how they have supported new business and renewal of existing business. And if these expenses are fixed or variable as a % premium.

- Then try to categorise each expense item according to their nature in following brackets:
 - Between fixed or variable
 - Between direct or indirect
 - By product / class of business
 - by function (type or timing of expense: initial, renewal, termination)
 - further sub-division (eg initial between sales / marketing and administration)
 - by what the expense is most closely linked to (or type of expense loading)
- Check if the current per-policy expense loadings, when summed up across all policies, are sufficient to cover the actual annual operating costs of the insurer.
- Expense inflation must be considered.
- The parameter values for expenses should reflect the expenses expected to be incurred in processing and subsequently administering the business to be written under the product being priced. [6]

ii) For the purpose of setting premium rates for new products:

- To set the expense loadings for the future period, the results from the historic expense analysis should be inflated at an appropriate rate from the mid-point of the historic observations to the mid-point of the future period for which the expense assumptions will apply. [½]
 - Rates of inflation will be partly related to prices and partly related to salary costs. [½]
 - The recent expense inflation experience of the company (or, if not available, of the industry) should be analysed to determine the basis for future projection. [½]
 - The expense assumptions may need to be adjusted for known changes e.g. changes in business mix, changes in underwriting/claims management procedures, changes in distribution channel etc. [1]
 - Expense loadings need to allow for the volume of existing business [½]
 - And the likely volumes of business to be sold in the future planning period. [½]
 - This is a particular area of uncertainty for a competitive product line. [½]
 - Then arrive at expense loadings as: [½]
 - New Business related – Fixed
 - New Business related – as a % Premium
 - New Business related – as a % of sum assured
 - Existing Business related – Fixed
 - Existing Business related – as a % Premium
 - Fund related expenses
 - Claims related expenses
 - Further, we need to adjust expense loadings due to competitive pressure. [½]
- [3]**

iii) Additional considerations when setting per policy expense assumptions for a CI product:

- The insurer is coming up with new product with no past experience. Hence, the expense loadings arrived from expense analysis based on endowment business must be adjusted to make it suitable for CI product.
- **New operating and development Cost:** [2½]
- Since the new CI product will be sold through online channel, hence the cost of setting up this channel will be incurred in addition to regular expenses.
- This may include contracting with web aggregators, setting up call centre, online website changes etc.
- Sales Training may be required for tele callers, which would incur additional cost.

- Product development costs for the CI product are significant e.g. effort is needed to derive incidence rates, develop policy literature and train staff.
- These development costs typically won't be included in the policy expense loadings but would be accounted for as development costs in the accounting period in which the work occurs.
- If existing permanent staff are working on the development then need to consider how their salary and salary-related costs will be treated after the development is completed e.g. if they return to regular work then their costs can be attributed to on-going/renewal costs.

Higher underwriting Cost:

[1½]

- The new product is with minimum 10 lakhs of sum assured and hence, the underwriting requirements shall be higher. However,
- Since this is a on line product, normally underwriting would be restricted to simplified health questionnaires and underwriting may largely automated
- There would be requirement to develop separate underwriting module for this product
- There could be requirements for health check-ups for some cases, if the answer is "no" to a particular health questionnaire., and a simple "accept" /"reject" approach may not work All this will result in underwriting costs which shall be included in per policy expense loadings while pricing this product.

Reinsurers:

[1]

The reinsurance structure for the product may be more complex than existing products hence cost of administration of reinsurance may need to be considered explicitly.

Higher claims Cost:

[1]

- CI product require more stringent claims underwriting and hence, increased claims cost shall be estimated.
- This per policy claim cost shall be accounted for while pricing this product.

Others:

[2]

- An appropriate method would be needed to split the overheads between CI and endowment business.
- E.g. split rent based on floor space occupied by each team, split per desks/headcounts allocated to each product line. This also depends on projected new business in CI product.
- Further, per policy expense loadings for endowment business will also get reduce if we split overhead cost.
- Inflation of expenses related to CI product could be different to the inflation related to endowment business.
- As some of the expense assumptions are set as a percentage of the premium then need to consider how the premiums increase in future in relation to the per policy expenses.

[7]

[16 Marks]

Solution 11:

i)

- The foremost risk is that the technical design and implementation process is not correct and accurate. This may result in non-achievement of objectives. [1]
- Economic Viability of the project: The objective of the project is to develop reusable satellite launch vehicles. This is an expensive project to manage it is critical that the net results shall give profits. The project carries a huge financial risk sufficient profits will not get generated.

[1]

- Technical risks [1½]
 - Inadequate and incomplete design
 - Errors in execution
 - Lack of specialized and skilled staff
- Other Financial risks [2]
 - Inflation- costs raising faster than expected
 - problems in raising the necessary finance initially / sponsor default
 - Private company- Limited access to capital
 - In accurate time estimates
 - Delayed deliveries
- Competitor risks- [1]
 - Competitors may be working on similar technology
 - Risk that they develop it faster or cheaper
- Security risks [1½]
 - Vandalism
 - terrorism
 - Crime, e.g. fraudulent activities by workers.
 - new technology prone to industrial espionage need for strengthening internal controls
- Physical risks [1]
 - Natural catastrophes
 - Workers safety
 - Environmental risks and poor disposal of toxic waste
- Underestimation of costs in the planning and construction phases, eg higher labour, material costs [½]
- Events leading to delays in the planning and the construction phases, eg strikes, problems [½]
- Delay in clearance with all government authorities [½]
- Overestimation of the usage of the RSLV after building: [1]
 - Cost constraints
 - Competitors developing even cheaper but similar technology
- Underestimation of the costs of running/maintaining the RSLV [½]

[10 Marks]**ii) Risk Mitigation:****[1 each]**

Type of Risk	Mitigation
Fault in the Technical design	A detailed research work needs to be done before project kicks off, consult global experts (ISRO, NASA etc.) to give their view on design and implementation process
Economic Viability of the project	• The method of determining the economic efficiency of reusable space-rocket systems must be studied in detail so that it results in net profits in the long run. • Cost vs benefit analysis is required with expert advice
Risk that developed reusable SLV s are null and void	• Possibilities for their effective application need to be studied so that reusable satellite launch vehicles can be used further • Significant testing and results proofs are needed

Underestimation of the cost of materials	Consult with mechanical experts, sub-contract to a third party, enter into forward agreements
Delays in planning and construction	• sub-contract to a third party, obtain insurance to protect against environmental disasters, • include clauses in the contractor's contracts that impose penalties if there are delays.
Underestimation of running costs	Research and estimate the costs incurred in other similar projects, consider future inflation also
Recurring refurbishment cost is more than expected	• Look at cost incurred by other similar projects and do cost benefit analysis for sustainability of the project
Fraud by employees and technical team	• Appropriate penalty clauses shall be mentioned in the contract which reduce / restrict frauds • If required, take appropriate insurance from global insurers

[8]

[18 Marks]
